CSC662 - MARKING SCHEME (mock paper, October 2026)

Confidence labels: computed / computed - convention unverified / placeholder.

PART A Q2a (2 marks) computed (solver validated on real papers)

```
Using Caesar Cipher technique, decrypt the following ciphertext.
KLCPJLYLABYULKHUKDPWLKVUL
(2 marks)
```

DEVICERETURNEDANDWIPEDONE
PART A Q2b (6 marks) computed - convention unverified (Playfair I/J + X padding pending slide check)

```
Encrypt the message below using Playfair encryption with the key REDFOX.
REVIEW ACCESS PERMISSIONS
(6 marks)
```

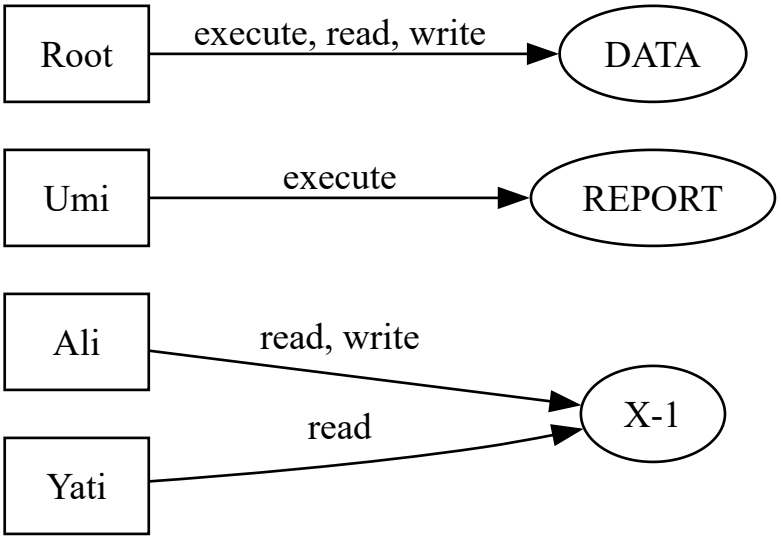
EDEPDVBGAFNCTQDEHKNCPLRTNC
PART A Q3b (6 marks) computed (consistency machine-verified)

```
Assume a computer system with the accounts: Ali, Yati, Umi, Root.

Ali can read, write X-1.
Yati can read X-1.
Umi can execute REPORT.
Root can execute, read, write DATA.

a) Write the access control matrix (ACM).
b) Write a set of access control lists (ACL).
c) Write a set of capability lists (CL).
(6 marks)
```

| Subject \ Object | X-1 | REPORT | DATA | |---|---|---|---| | Ali | readwrite | - | - | | Yati | read | - | - | | Umi | - | execute | - | | Root | - | - | executereadwrite | **ACLs (per object):** - X-1 -> Ali: readwrite; Yati: read - REPORT -> Umi: execute - DATA -> Root: executereadwrite **Capability lists (per subject):** - Ali -> X-1: readwrite - Yati -> X-1: read - Umi -> REPORT: execute - Root -> DATA: executereadwrite **Diagram (Graphviz dot):** ```dot digraph ACM { rankdir=LR; node [shape=box]; "Ali"; "Yati"; "Umi"; "Root"; "X-1" [shape=ellipse]; "REPORT" [shape=ellipse]; "DATA" [shape=ellipse]; "Ali" -> "X-1" [label="read, write"]; "Root" -> "DATA" [label="execute, read, write"]; "Umi" -> "REPORT" [label="execute"]; "Yati" -> "X-1" [label="read"]; } ```



PART A Q4a (2 marks) computed (solver validated on real papers)

```
Using Caesar Cipher technique, decrypt the following ciphertext.
AYHPULTWSVFLLZVUZLJBYPAF
(2 marks)
```

TRAINEMPLOYEESONSECURITY
PART A Q4b (6 marks) computed - convention unverified (Playfair I/J + X padding pending slide check)

Encrypt the message below using Playfair encryption with the key REDFOX.
CONTAINER SECURITY CHECKS
(6 marks)

GFPNIPPRFNFARXMPFLIRBLNC

PART A Q5b (6 marks) - Q5b no computed question available in this slot

All remaining slots (theory placeholders, 72 marks) placeholder - sourced-theory module not built yet